

A working model of behavior leakage: a constant write, gated by context similarity

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1 Setup

Leakage: a behavior trained into one context (system prompt, persona, ICL prefix) also appears under other, untrained contexts. Measured as the trained–base change in expression under a bystander context, on-policy, with a behavior-appropriate metric: marker-token log-probability at the end of the bystander’s own response (#502/#489); judge-scored agreement for sycophancy (#509); misaligned-and-coherent rate for EM (#521); taught-fact emission for fact implants (#500).

1.1 The general problem, before any model

The response computation sees a context only through its *KV cache*. So, with no model at all: any predictor of leakage is a function of the base model’s KV cache, and any gate training could implement is an embedding of it. Two questions, posed at this level first:

- **Prediction.** How far is fine-tuning generalization predictable from base-model quantities, before training?
- **Control (the best-effort question).** Take the *most behavior-discriminatory* embedding of contexts — nonlinear, any function of the KV cache — suppose training could gate on it, and ask whether leakage persists.
 - *Inevitable* $\Rightarrow$ name why. Lazy candidate: a small edit can only gate on directions the base model represents $\approx$ linearly; inevitable leakage = best discriminator – best linear gate on frozen features.
 - *Evitable* $\Rightarrow$ exhibit the edit. Linear candidate: the multi-pair contrastive edit (A2’s relaxation); protection $\approx$ span of the trained negatives, so off-panel leakage measures best effort’s reach.
 - Linear case = geometry: evitable iff the source key is linearly separable from the bystander span; contrastive negatives estimate the discriminant. Off-panel generalization of a K -negative discriminant: empirical, not a theorem.

The model answers both by specialization: four assumptions, general $\rightarrow$ specific, each buying a cheaper predictor, each with a relaxation. A miss at rung k means climb to rung $k-1$, not refutation; only the conjunction is refutable (exact gate + measured write failing to rank a held-out panel leaves no rung standing).

2 The assumption ladder

The objects.

- Pretraining leaves a dictionary of behaviors b , each carried by a residual-stream direction. Two roles, named separately: *adding* a steering vector d_b elicits b ; *projecting* onto a read-out r_b measures it.
- The state under a context decomposes over the dictionary: $f(c) \approx \sum_b \alpha_b(c) d_b + \text{error}$. The pretrained associations $c \rightarrow b$ ARE the coefficients; “the assistant is not a villain” is a statement about weights, not the dictionary. (Idealization: directions are neither orthogonal nor a basis; literal for the marker at read-out, empirical for EM, an open bet elsewhere.)
- Two typed functions: $f : c \mapsto f(c)$ — *what training edits*; the table $A[b, c] = \langle r_b, f(c) \rangle$ — *what experiments measure*.

A0 (lazy regime): the update is small in weight and feature space

- **Statement:** fine-tuning acts to first order — it re-binds contexts to states the base model already represents, builds no new features. Edits the weights $\alpha_b(\cdot)$, never the dictionary. Corollary: implantable $\Leftrightarrow$ elicitable. Small in weights $\neq$ small in behavior: read-outs amplify (discrete emission flips; EM via nine rank-1 adapters [7]).
- **Buys:** the most general predictive form, $\Delta f(c) \approx K(c, c') \cdot w$ — w the write (correction achieved at the source), K the empirical tangent kernel (strictly a cross-kernel), gate $g(c) = K(c, c')/K(c', c')$. Every rung below is a cheaper K .
- **Support:**
 - [15] — small updates move a network linearly in function space; the tangent kernel is this rung’s first-order object.
 - [16] — LM fine-tuning empirically operates in this kernel regime; the license at 7B scale.
 - [9] — narrow fine-tunes leave constant, linearly readable activation traces: a fixed write, not new circuitry.
 - [8] — narrow fine-tune effects reproduce as a near-constant steering write.
 - #514 — LoRA vs full FT leak identically at matched strength (+0.00 nat): parameterization-irrelevance, as first order requires.
 - #538 — $3\times$ implant depth, edit geometry unchanged: dose moves magnitude, not features.
- **On failure:** no rung above. Rich regime (features move) kills every base-model predictor (§4); the EM predictor failure looks like it ($\rho = 0.09$, #458; assistant-axis rotation 38–53°, discrimination collapse, #125). Lazy-vs-rich is measurable, not assumed: the two-channel decomposition (§3).

A1 (local linearity): the edit acts at one layer, where the map is linear

- **Statement:** at one layer $f(c) \approx Mv_c$, and the edit acts there. v_c = pooled activations over a probe set — the cheapest KV-cache function (§1.1); what the gate later misses traces to what the pooling threw away.
- **Buys:** the kernel collapses to an inner product from base forward passes alone: $g(c) = \langle v_{c'}, v_c \rangle / \|v_{c'}\|^2$ — asymmetric: $c' \rightarrow c$ scales as $\cos \theta \cdot \|v_c\| / \|v_{c'}\|$ (P2).
- **Support:**
 - [10] — linear representation hypothesis: features occupy $\approx$ linear subspaces; the prior that makes an inner product the natural similarity.
 - #502/#509 — proxy ladder (cosine $\rightarrow$ distributional $\rightarrow$ exact kernel) gains little (0.79 vs 0.76 marker; 0.50 vs 0.49 sycophancy): the linear rung is not yet the bottleneck.

- P5 empirics — predictor quality peaks at behavior-specific layers (marker 19–24, sycophancy 1–8); consistent, pending independent localization.
- **On failure:** climb to A0’s kernel. Under A3’s router relaxation the gate is not a similarity at all: attention-shaped, near-binary for discrete triggers, estimated from samples.

A2 (minimal edit): the smallest weight change that fits the new pair

- **Statement:** the update is the rank-one key→value write of linear associative memories: one key (the source context), one written direction.
- **Buys:** (i) with A1, the factorization, eq. (1): write $\times$ gate — *what* leaks decoupled from *who* leaks; (ii) the weight-space face: $\Delta W \approx \text{write} \otimes \text{key}$, so SVD of a trained adapter returns both (P0); (iii) the in-context extension: an ICL prefix acts as a transient rank-one patch — same write $\times$ gate, different persistence (P7).
- **Support:**
 - [11, 12] — linear associative memories: rank-one outer product = the minimal store of one key→value pair; the rung’s mathematical core.
 - [13] — ROME: a rank-one edit to one MLP layer rewrites a factual association.
 - [1] — weights as read/write pairs, rank-one outer product the atomic unit: the weight-space vocabulary.
 - [17] — LoRA: adaptation succeeds under an explicit low-rank constraint — needed updates have small effective rank.
 - [2] — rank-1 LoRAs carry interpretable, causal signals: rank one suffices to implement a behavior change.
 - [18, 19] — a transformer block provably converts an ICL prefix into a transient rank-one patch: the extension’s formal basis.
 - [20] (>85% of in-distribution FT corrections recovered in-context), [23] (ICL anticipates FT generalization failures), [24] (ICL compresses to one direction) — the behavioral form, which is all P7 claims.
 - [21, 22] — strict ICL=GD equivalence fails under scrutiny: why P7 claims only the behavioral form.
 - #489 — the similarity gate already ranks leakage from ICL sources.
- **On failure:** *low-rank* — write spans a subspace; P1 weakens to “shifts span r dimensions,” the SVD in use measures r . *Multi-pair* — the minimal edit solves a kernel system whose off-diagonals are the contrastive negatives, subtracting the common mode of near-parallel keys ($\cos \approx 0.92$): why positive-only implants leak near-uniformly and contrastive ones localize (99.6% vs 11.7%, #247/#329). Early stopping spectrally filters toward raw similarity (untested).

A3 (linear read-out): each behavior is measured by projection onto one direction

- **Statement:** behavior b is observable as the projection onto r_b (unembedding row for the marker; probe or contrastive direction otherwise). $r_b \neq d_b \neq v_b$; collapsing them assumes the downstream network is the identity on the write.
- **Buys:** the scalar observable $\Delta z_b(c) = \langle r_b, w \rangle g(c)$ — one slope per behavior $\times$ gate; makes the write measurable. Cross-behavior transfer: one write, many read-outs (P6). Measurement note: the softmax is over *tokens*, not behaviors — behaviors compete at emission (marker vs end-of-turn), log-probs compress near ceiling, so logits are stored alongside. Marker:

expression = one logit (the clean testbed); sycophancy: sequence-level, judge-scored.

- **Support:**

- [3] — adding one residual direction elicits the behavior without optimization: steering directions exist.
- [4] — contrastive activation addition steers Llama 2 across behaviors, one direction each.
- [5] — persona vectors: single-direction projections track traits before generation: read-outs exist, predictive.
- [6] — one persona feature predicts and controls EM.
- [7] — independent EM fine-tunes converge to one linear direction; ablating it removes the behavior.

- **On failure (router):** the write need not copy the behavior direction — a query shift can recruit attention that fetches the behavior from the context’s own values [8]. Cleanest case: a rank-1 EM adapter, write $\cos 0.04$ to the misalignment direction, yet 98% of its effect ablates along it downstream [7]. Behavior stays one direction at read-out; P1 can fail by mechanism.

3 The formula at the bottom of the ladder

Training drags the source state $v_{b'} = f(c')$ toward v_b , landing at $v_{b''}$ (early stopping undershoots, so the write is measured, not assumed). A2 with A1 — minimize $\|\Delta M\|_F$ s.t. $(M + \Delta M) v_{c'} = v_{b''}$ — gives

$$\Delta f(c) = \underbrace{(v_{b''} - v_{b'})}_{\text{write: } \textit{what} \text{ leaks}} \cdot \underbrace{\frac{\langle v_{c'}, v_c \rangle}{\|v_{c'}\|^2}}_{\text{gate: } \textit{who} \text{ leaks}} \quad \text{i.e.} \quad f'(c) = \left(M + \frac{(v_{b''} - v_{b'}) v_{c'}^\top}{\|v_{c'}\|^2} \right) v_c. \quad (1)$$

Division of labor: A2 makes the edit rank-one, A1 makes the gate an inner product, A2’s weight face makes it visible in ΔW , A3 makes the write measurable ($\Delta z_b(c) = \langle r_b, w \rangle g(c)$).

One rank-one nudge to the association table. Eq. (1) compresses to $\Delta A[b, c] = \langle r_b, w \rangle \cdot g(c)$. The intended edit is one cell (b, c') ; the minimal edit smears along both axes:

- down the columns, every context moves $\propto$ its gate — *leakage* (P2);
- across the rows, every behavior moves $\propto$ its slope $\langle r_b, w \rangle$ — *cross-behavior transfer* (P6);
- so leakage = off-target table movement: $\alpha_b(c)$ changing for pairs the fine-tune never targeted — generalization nobody asked for.

The write is not the read-out.

- Collapsing v_b and r_b assumes the downstream network is the identity on the write; kept separate, it can rotate, amplify, or route. The anomalies live in that gap — all of the form “write $\perp$ read-out, yet the behavior moved.”
- Three on record: prompted marker direction vs realized shift ($\cos \approx -0.03$, #521; P8); the routing adapter (A3); the P1 split (§6: marker and EM arms identical at read-out, differ in rank-one-ness of the shift matrix).
- Not oversold: ranking who leaks needs no v_b ($\Delta z_b = s_b g(c)$, s_b fitted). The two objects earn their keep in the vector-valued predictions (P1, P3', P6, P8).

The edit has two channels. With $f(c) = Mv_c$: $\Delta f(c) = \underbrace{\Delta M v_c}_{\text{re-bind map}} + \underbrace{M \Delta v_c}_{\text{move representation}} + \Delta M \Delta v_c$.

- Nothing confines the update to M (an all-linear adapter touches every layer below the read). Eq. (1) is channel 1 alone; geometry persistence (§4) = “channel 2 vanishes.”
- Both outcomes on record: $\Delta v_c \approx 0$ at $3\times$ depth (#538); assistant-axis rotation $38\text{--}53^\circ$ + discrimination collapse under EM (#125).
- Measurable on stored artifacts (#521/#551 tensors): Δv_c directly, the $\Delta M v_c$ share by fitting on frozen base summaries, the rest = moved representations. Cheapest candidate account of the P1 tension: marker and EM arms may use different channels.

Contexts encode behaviors.

- Contexts and behaviors are both activation states: a persona prompt’s projection profile $\{\langle r_b, f(c) \rangle\}_b$ is the behavior bundle it elicits without training. Used in three places: source residual (P3), bystander prior (P3), the estimator ladder (§4).
- Not licensed: treating behavior and context as separate axes. The gate mixes prompt similarity with behavior-bundle overlap; an instruction context can encode the behavior while sitting far from the source — the regime where geometry collapsed and the base prior carried the signal (#532).
- Open test: vary behavior overlap at fixed context similarity. Candidate gate term: similarity conditioned on the taught material (JS over the taught answers, teacher-forced) — close = agreeing about b ; its scalar limit re-derives the base-prior predictor (P3, #532) as a gate term.

Cast of characters.

Object	Type	Role	Moved by training?
v_c	vector	context key (input side), read while processing c	no (base-model object)
f	map	contexts $\rightarrow$ states; holds the associations	yes — the only thing edited
$f(c)$	state	where the model sits under c	moves when f moves
$\alpha_b(c)$	scalar	association strength of $c \rightarrow b$ (the weights)	moves (this <i>is</i> the edit)
d_b	direction	steering: add to elicit b	no (dictionary)
r_b	direction	read-out: project to measure b	no (and directly observable)
v_b	state	where the model is when performing b (target)	no (dose-independent)
$v_{b'}, v_{b''}$	states	source state before / achieved after training	$v_{b''}$ is the outcome
$w = v_{b''} - v_{b'}$	vector	the write: <i>what</i> leaks	created by training
$g(c)$	scalar	the gate: <i>who</i> leaks	no (computed from base v_c)

4 Reading everything from the base model

v_b is the *post-training* state — unobservable before training — so estimating it from the base model is part of the proposal.

- **Estimators** (decreasing fidelity): (i) teacher-forced reads over the prospective training data’s completions; (ii) in-context demonstrations from that data (the variant run so far); (iii) description conditioning. All are f at a behavior-eliciting context; validated against the realized $v_{b''}$ and forward-predicted leakage (P7–P8). Standing warning: $\cos \approx -0.03$ between prompted marker direction and realized shift (#521). Constructions (layer, position, aggregation) are swept, not fixed.

- **Prompting stands in for fine-tuning at three points**, which fail separately:
 - *Mechanism* — same write $\times$ gate at different persistence (A2’s extension); equivalence open, so P7 claims only the behavioral form.
 - *State identity* — prompted-performance state = fine-tuned state. Most exposed (P8; the $\cos \approx -0.03$ warning); teacher-forced is the principled estimator — same forward passes the gradients come from.
 - *Geometry persistence* — base v_c still measures the trained model: A0 restated; the one substitution no better estimator repairs.
- With the realized $v_{b''}$, eq. (1) is descriptive and needs none of the three; they are the price of prediction *before* training — the program’s point. Most of §6 implicitly tests them.

5 Predictions

P0 (weights decompose).

Top singular vectors of ΔW = key and write; under contrastive training the key moves toward a source-minus-negatives direction with dose.

P1 (one write direction).

Per-bystander shifts at the read-out all point along $(v_{b''} - v_{b'})$: shift-matrix SVD near rank-one, top component matching the behavior delta.

P2 (gate = base-model context similarity).

A fixed kernel on base context summaries sets who leaks, transferring across held-out panels. On A1’s rung: the dot product, hence asymmetric ($c' \rightarrow c \propto \cos \theta \cdot \|v_c\|/\|v_{c'}\|$).

P3 (the behavior’s prior matters).

Source side: write scales with $\|v_{b''} - v_{b'}\|$ — a source already performing b leaks little despite high similarity. Bystander side: prior enters via read-out compression (measurement) and possibly the gate (mechanism); separable in logit space only with the exact gate.

P3' (source prior shapes the write’s direction, not just size).

High-prior source $\Rightarrow$ residual concentrates in source-specific subspace: prior predicts the write’s *common-mode fraction*, not its norm. #541: high-prior teachers express most on themselves (97/94% vs 72%) while leaking least (“not weaker, narrower”) — impossible under magnitude-only P3.

P4 (additivity).

Joint two-implant training $\approx$ sum of singletons; deviations set by source mutual similarity (activation-space task arithmetic [14]).

P5 (layer locality).

The gate predicts best at the layers computing the behavior; needs independent localization (predictor-quality scoring is circular). Behavior-specific: marker 19–24, sycophancy 1–8.

P6 (cross-behavior transfer).

One write moves other behaviors $\propto$ its projection onto their read-outs.

P7 (ICL proxy).

Base-model in-context behavior on the training mix forward-predicts the post-training leakage map (A2’s extension).

P8 (the estimator is valid).

The three v_b estimators agree with each other and the realized $v_{b''}$. If not: formula predictions fail in a way the realized $v_{b''}$ repairs — estimation failed, not the update rule.

No single miss refutes the model (each rung has its relaxation, §2); the conjunction does (§1.1).

6 Experiments

	Experiment	Status
P0	SVD stored adapters; compare singular vectors to keys and measured writes; cross-compare [8]’s per-trigger backdoor vectors	Not yet run; free on existing artifacts.
P1	SVD per-context activation shifts, marker implant vs EM SFT (#521, controls #551)	Split: EM shifts all 14 contexts along one seed-stable direction ($\cos 0.96\text{--}0.98$); the marker implant does not, its trained context <i>least</i> aligned. #552 (benign-SFT control) and #561 (positive-only retrain) running.
P2	Correlate base-model similarity with measured leakage, source $\times$ bystander panels (#502, #489, #509)	Ranks marker leakage ($ \rho \approx 0.76\text{--}0.79$, layers 19–24), extends to ICL/multi-turn — but only ordinarily (out-of-fold $R^2 \approx 0$); no cross-panel transfer (#553); fact arm reversed sign while the base prior predicted correctly (#500) — an in-regime miss.
P3	Instructed-source break case; regress logit shifts on gate proxy + bystander base logit (#500, #531, #532)	The bystander’s own prior is the most consistently supported predictor and beats geometry on instruction contexts (#532); measurement-vs-mechanism split awaits the exact gate.
P3’	Project per-context shifts onto shared vs source-specific components; regress common-mode fraction on source prior (#541’s nine fact adapters; #518’s refusal/EM adapters)	Not yet run; activation extraction on existing adapters only.
P4	Joint two-source training vs singleton sum (#520, #527, #538)	Not yet diagnostic: singleton shifts are themselves near rank-one, so the additivity cosine (0.99) is trivially high at every tested dose.
P5	Layer sweep of predictor quality per behavior + independent localization	Marker best late (19–24), sycophancy early (1–8); consistent but unconfirmed without the localization step.
P6	Behavior-generalization testbed (#545); context testbed (#537); formula-as-predictor (#510)	#537 running; #545, #510 queued.
P7	ICL-vs-FT equivalence, persona-framed (#491): same K examples in-context vs fine-tuned; ICL-conditioned base responses as leakage predictor	Queued; the gate already extends to ICL sources (#489).
P8	Estimator bake-off: three v_b estimators vs realized post-training states on stored adapters (#493 engine, #521 tensors)	Not yet run; free on existing artifacts. Warning on record: $\cos \approx -0.03$ (#521).

Headline experiment — dose titration: implant at a non-saturated anchor (#448); measure expression change, activation shift, and gate proxy per held-out context, on-policy (#432→#456), across the 10–25-step cliff (#139), ≥ 3 seeds. Predicted: near rank-one, similarity-graded shifts at low dose; rotation and prediction collapse across the cliff. The P1 split complicates the low-dose

half, so the titration doubles as its diagnosis.

The regime tension. A0 is where the model lives or dies. Current tension: the marker arm passes the gate test but fails the one-direction test; EM does the reverse. #552/#561 decide rig vs regime; the per-arm channel shares (§3) read the same question off stored artifacts. The durable question under every version: how far is fine-tuning generalization predictable from base-model activation geometry, before training?

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